# One timetable, fewer footsteps, no wasted periods

A constraint-certified, auto-scheduling system for 626 students, 54 teachers and 63 rooms — with a measured, stres improvement over the current way of building the grid.

## 1 What is wrong today

From the one real G11-AP week we have, and a catalog audit:

- **Students carry the walking.** Every teacher owns a room, so a student changes rooms ~**21×/week** (488 s of walking on the observed week).
- **Slots are filled with no objective**: stranded free periods between classes (2.8/week), hard subjects late in the day.
- **Period length is ignored.** Periods run 60/40/40/45/55/40/40 min, so swapping a Math lesson from P1 to P2 silently removes 20 min — breaking the one rule the brief names: *hours must not change*. 110 of 270 sections drift out of band in a typical status-quo grid.
- **Nothing is verified** and a room loss is repaired by hand.

> **Honest scope.** We have one real week. The "current system" here is an emulator calibrated to it (~20 vs 21 observed room changes), run 24 times to get a spread.

## 4 The mathematics

### LOADS: CATALOG CREDIT → WEEKLY SLOTS

$$n_c = \left\lfloor k\,\kappa_c + \tfrac{1}{2} \right\rfloor, \qquad k = \frac{\sum_i \kappa_i\, o_i}{\sum_i \kappa_i^2} \approx 0.873$$

$\kappa_c$ catalog credit, $o_i$ slots observed on the real week; swept $k \in [0.83, 1]$.

### HARD CONSTRAINTS (DECISION $Z_{G,S} \in \{0,1\}$)

$$\sum_s z_{g,s} = n_g \quad \sum_{g \in \sigma} z_{g,s} \le 1 \quad \sum_{g \in T_t} z_{g,s} \le 1 \quad \sum_{g:\, a_g \ge a} z_{g,s} \le R_\kappa(a)$$

students $\sigma$, teacher $t$, rooms of kind $\kappa$ with area $\ge a$ (nested ⇒ Hall's condition is exact).

### CONTACT-HOURS INVARIANT (MINUTES, NOT SLOTS)

$$\left| \sum_s \mu_s z_{g,s} - n_g \bar{\mu} \right| \le \max\left(\tau\, n_g \bar{\mu},\ 9\right), \qquad \tau = 0.08$$

### STAGE-1 OBJECTIVE

$$\min\ w_{\text{gap}} \sum_\sigma c_\sigma \Gamma_\sigma + w_h \sum_\sigma c_\sigma H_\sigma + w_{sp} \sum_g |g|\, \text{rep}_g + w_m \sum_{g\ \text{hard}} |g| \sum_s (p_s - 1) z_{g,s} + w$$

$\Gamma$ idle gaps, $H$ days with ≥6 classes, rep same-day repeats, $c_\sigma$ students in signature $\sigma$.

### STAGE-2 OBJECTIVE: PERSON-SECONDS

$$\min_r \sum_{(a,b) \in \mathcal{L}_{\text{stud}}} n_{ab}\, \tau(r_a, r_b) \ + \ \lambda \sum_{(a,b) \in \mathcal{L}_{\text{teach}}} \tau(r_a, r_b)$$

### COMPOSITE SCORE AND FAIRNESS

$$J = \bar{\ell}_{\text{student}} + \alpha\, \overline{L}_t + \beta\, \text{CVaR}_q(L), \qquad \text{CVaR}_q = \frac{1}{\lfloor (1-q)n \rfloor} \sum_{\text{worst}} L_i$$

### ROBUSTNESS

$$\mathbb{E}[\Delta] = \sum_\xi p_\xi\, \Delta(\xi), \quad \Delta = \text{moved lessons} + \text{room changes} + \text{uncovered perio}$$

## 2 What we optimise & who it affects

| ASPECT | WHO | WHY IT MATTERS |
|---|---|---|
| Room changes & walking | 626 students | transit time, late arrivals, congestion on stairs |
| Idle periods between classes | students, teachers | stranded time that cannot be used |
| Hard subjects in the morning | students | attention; mean period 3.7→3.3 |
| Equal minutes per course | everyone | the "hours unchanged" invariant |
| Teacher gaps, fatigue cap | 54 teachers | fair, humane workload |
| Repairability | admin | closures, absences, surges |

Students stay, teachers walk: one teacher-second counts $\lambda$ of a student-second (swept, see Pareto).

## 3 Method: four stages

**0 Sectioning.** Catalog rules → offered *pathways*; a pathway = one class with an identical timetable (kills cross-class clash cliques).

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**1 Slots (CP-SAT).** Boolean $z_{g,s}$ per section×slot — linear, not $O(M^2)$ pairwise.

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**2 Rooms.** Cohort home rooms → per-slot min-cost matching → exact person-seconds simulated annealing.

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**3 Certify.** A separate checker re-derives 10 rules from the finished grid; mutation tests prove it can fail.

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**4 Repair.** Substitute → re-room → re-slot only the affected sections, all else frozen.

Constraints from the course catalog · building model: 6 floors, 103 rooms, stair-aware walking times · seed-reproducible, `python -m tt.run`